

roomscan: submission summary

Repository: https://github.com/shoraaz/Floor_plan_lidar - one command per capture: `uv run roomscan run <capture> --out <dir>`

Deliverables

#	Deliverable	Location in repo	Status
1	Compliance matrix	docs/COMPLIANCE_MATRIX.yaml	done (15 done / 6 partial / 4 not done, each with reason)
2	Capture route + device matrix	docs/CAPTURE_PROTOCOL.md (Route 2, stock tools), docs/DEVICE_MATRIX.md	done
3	Repo, README, one command per capture	README.md, src/roomscan/cli.py	done; uv-based install
4	Reproduction bundle	reproduction/run_final.ps1, get_arkitscenes.ps1, fixloop.ps1; deterministic caches; live path runs	done (Windows PowerShell scripts)
5	Benchmark report	benchmark/results/FINAL_RESULTS.md (gates, repeatability, tier tables, timing)	partial: no head-to-head (no device)
6	Fix loop bundle	fixloop/ (declaration, before/after, diff, post-mortem), tags fixloop-before / fixloop-after	done (gate not passed, honest post-mortem)
7	Technical report (6 pages)	docs/REPORT.md (reproduced below)	done
8	Raw benchmark data	own: ARKitScenes venue 470350 (fetched by script, licence prevents re-hosting); sample: organisers' captures	partial: no app exports

Data note. No LiDAR phone or room was available, so the own benchmark uses Apple ARKitScenes (CC BY-NC-SA 4.0): three iPad-LiDAR captures of one venue with Faro laser-scanner ground truth, converted to the Stray Scanner format so the unchanged pipeline runs on them. All numbers are also reported on the organisers' sample data.

Headline (honest). LiDAR tier: metric point cloud (0.8% scale vs laser); wall lengths median 5.0 cm on walls with unambiguous laser ground truth, 17.8 cm over all scored walls; no centimetre gate passes. Video and photo tiers run end to end but fail their gates and flag themselves UNRELIABLE rather than emit confident garbage.

Gate table

gate	target	measured	status
Wall lengths (LiDAR) vs laser	<= 1 cm or 0.5% (repeat gate used as accuracy proxy)	22 walls scored; median	err
Opening widths	<= 2 cm on >= 85%, missed/phantom = miss	7 doors scored; median	err
Ceiling height	<= 1.5 cm; spread <= 1 cm	laser GT 2.309, 2.311, 2.268, 2.267 m; ours reported in 2 rooms (ceiling not observed by the iPad sweeps -> withheld)	NOT SCORED
Repeatability (LiDAR)	<= 1 cm or 0.5% per wall	own a/b 0/8, own a/c 0/4, sample 4/24	FAIL
Drift accountability	method + on/off footprint ablation	pose graph + ICP-verified revisit loops, ON by default; ablation in drift_ablation_*.log (no measured benefit on sample data)	PARTIAL
Photo-tier whole-property stitch	one stitched plan, correct adjacency, footprint +/-8%	see tier table: collapses / flagged UNRELIABLE	FAIL
Video tier	walls +/-3%	see tier table	FAIL
Calibration / no confident garbage	intervals honest at every tier	LiDAR interval coverage vs laser: 100%	PARTIAL

Own benchmark: LiDAR tier vs Faro laser ground truth

capture	walls scored	median abs err (cm)	within 2 cm	within 5 cm	doors scored
ark470350_a	12/20	19.38	0	2	4
ark470350_b	4/8	44.44	0	0	0

capture	walls scored	median abs err (cm)	within 2 cm	within 5 cm	doors scored
ark470350_c	6/12	4.98	2	4	3

roomscan: technical report

Applied AI Engineer case study. Every number here is regenerable: `reproduction/get_arkitscenes.ps1` (own benchmark data), `reproduction/run_final.ps1` (all tiers, all captures, scoring), `scripts/final_tables.py` (benchmark/results/FINAL_RESULTS.md), `reproduction/fixloop.ps1` (fix loop). Short version: the LiDAR tier is a complete, metrically sound pipeline that **does not yet meet the centimetre gates**; the video and photo tiers run end to end but are far from their gates and say so in their output. All gate results are in sec. 4.

1. Data and the one substitution

- **Sample data** (organisers): 3 Stray Scanner LiDAR captures, two of the same flat. No ground truth supplied.
- **Own benchmark**: no LiDAR phone or room was available to me, so I used **Apple ARKitScenes** venue 470350 (CC BY-NC-SA 4.0): three iPad-LiDAR captures of the same space plus **Faro laser-scanner point clouds** as ground truth. `scripts/arkitscenes_to_stray.py` converts each capture to the exact Stray export format, so the unchanged `roomscan run` processes it (also a rehearsal of the walk-in format). Two conventions were verified on data, not assumed: ARKitScenes trajectories are world-to-camera in a **z-up** world (camera height std 0.30 m along z; floor peak along z), rotated to the y-up frame Stray uses. Not covered by this substitution: staged damage, a 3+-room-plus-connector layout, my own capture protocol, the consumer-app head-to-head.

2. Architecture

```
LiDAR (Stray: depth + confidence + ARKit poses + K) -> pose-graph drift correction -> fuse
Video (RGB clip only) -> DA3-LARGE-1.1 chunks, per-chunk metric scale (DA3METRIC) -> fuse
Photo (room folders) -> one joint DA3 pass over all photos + metric scale -> fuse
all three -> metric, y-up point cloud + camera path + ray endpoints
shared backend: floor/ceiling planes -> Manhattan yaw (normals + 0.02 deg wall-sharpness search)
-> wall map in a 0.95-1.6 m band -> doorway closure -> wall extension through unobserved space
-> rooms = wall-enclosed regions -> each side snapped to the room-side wall face -> ceiling per room
-> doors (doorways joining two rooms; width from door-frame faces) -> intervals -> plausibility gate
-> stitch checks -> plan.json (schema-validated before writing) + plan.svg
```

Design decisions I would defend live: - **One backend, three front ends**. Room logic exists once; tiers differ only in how a metric cloud is made and in interval width. Cost: a weak front end cannot be rescued downstream (that is what happened to video/photo). - **Structure, not coverage, defines rooms**. Rooms are wall-enclosed regions, with walls extended through unobserved space (ROSE2-style); using observed space made room extent depend on where the phone pointed. - **Wall band 0.95-1.6 m**. Above tables, beds and counters, and covered by any protocol-following walk. One sample walk was aimed low (~5% of points above 1.6 m); a 1.1-2.0 m band broke its walls entirely. - **Never guess**. Ceiling is `null` when not observed; empty reconstructions produce a valid, empty, flagged plan; non-LiDAR plans with an implausible floor-to-ceiling distance (outside 2.1-4.2 m) are marked UNRELIABLE. - **Verify conventions on data**. Stray poses are camera-to-world, OpenCV camera, ARKit y-up world (floor peak 1.4 m below the camera only under this reading). DA3 extrinsics are world-to-camera (5-17 cm ATE per chunk vs 21-26 cm the other way). DA3METRIC depth in metres = output x focal / 300 (LiDAR ratio 1.37-1.59 vs 1.40).

3. Tiers and device matrix

Tier	Input	Runs on (processing)	Honest accuracy on our data
LiDAR	Stray Scanner export (iPhone/iPad Pro)	CPU; 18-32 s per capture	metric cloud (0.8% scale vs laser); walls median 5.0 cm where GT is unambiguous, 17.8 cm over all
Video	any clip (iPhone 15+)	CUDA GPU, 8 GB; 6 min cold for 90 s of video	footprint -3% to -90%; trajectory error ~5% of path
Photo	2-8 stills per room folder	CUDA GPU; 30-130 s	collapses to 0-1 rooms; flagged UNRELIABLE

4. Results

Gate table (benchmark/results/FINAL_RESULTS.md):

Gate	Target	Measured	Status
Walls (LiDAR) vs laser	1 cm / 0.5%	all 22 walls: median 17.8 cm, 2 within 2 cm; 6 clean-GT walls: median 5.0 cm , 0 within 2 cm	FAIL
Opening widths	≤ 2 cm on $\geq 85\%$	7 doors scored: median 11.5 cm; 0 within 2 cm	FAIL
Ceiling height	≤ 1.5 cm	laser 2.27-2.31 m; iPad sweeps rarely saw the ceiling -> withheld in most rooms	NOT SCORED
Repeatability	1 cm / 0.5% per wall	own a/b 0/8, a/c 0/4, sample 4/24; raw surfaces agree to 0.69-0.95 cm	FAIL
Drift	method + ablation	pose graph with verified loops, ablated (sec. 5)	PARTIAL
Photo whole-property stitch	+/-8%, correct adjacency	does not stitch	FAIL
Video walls	+/-3%	not met	FAIL
Calibration	honest intervals	LiDAR 91% coverage vs laser, video/photo 100% vs LiDAR (in-sample)	PARTIAL
Head-to-head vs consumer app	beat/tie $\geq 70\%$	not run: no device	NOT DONE

Where the LiDAR error comes from (laser, `scripts/laser_gt.py`, `scripts/diagnostics/scale_check.py`). The fused cloud is metrically right: similarity ICP onto the laser scan gives scale 1.008 and 1.9 cm RMSE. The error is in which surface a room side snaps to. Per-side comparison against laser wall faces (measured above furniture height): several sides land within 1 cm (-0.6, -0.7, -0.9, -1.2 cm), many land **18-24 cm outside, i.e. on the far face of the wall**. A fix for one cause (rectilinear outlines including the wall thickness) moved room1's worst sides from +23.8/+22.7 cm to -6.7/+2.2 cm. Listing every laser surface near each side (`laser_gt.py --peaks`) separates the remaining ~20 cm cases into two kinds: (a) **our side sits where the laser has no surface at all**, i.e. a virtual wall created by doorway closure or wall extension, which snapping cannot fix; (b) **the laser shows a 25-50 cm band of clutter, not a plane**, so the ground-truth face is itself ill-defined. The scorer therefore reports a clean-GT subset (one sharp laser surface, ≤ 5 cm wide with no rival, on both bounding walls): **median 5.0 cm over 6 walls**, against 17.8 cm over all 22. I tried choosing the room-side face from ray-cast free space (rays stop at the inner face); it made capture a worse (median 20.9 -> 25.9 cm, because it also moved room splits) and was not shipped (`scripts/diagnostics/gt_quick.ps1` reproduces both). The scorer handles rectangular rooms only.

Repeatability. Raw wall surfaces of two captures agree to 0.69 cm (own) and 0.95 cm (sample) after registration; footprints to 1.4% (own a/b). Per-wall agreement fails because the same space is partitioned or snapped differently per capture (sec. 6).

Video and photo (`benchmark/results/video/`). Within one 24-frame chunk DA3 is good to 5-17 cm over 3 m. Chaining chunks is the weak link: v1 chained relative scale and compounded it (scale off 86-91% after 54-60 m); v2 anchors each chunk to metric depth independently, which removed the blow-up but not the shape error (1.5-4.5 m jumps at chunk joins, ATE ~3 m on 98 m). COLMAP SfM registered only 13-21% of keyframes on these low-texture walls and is kept only as a fallback. The photo tier's joint pass does not hold rooms together; on the own capture it recovered no rooms and emits an empty, flagged plan instead of a guess.

Timing (RTX 5050 laptop): LiDAR 18-32 s; video 39-109 s with cached DA3, 380 s cold for a 90 s clip; photo 32-132 s.

5. Drift handling and ablation

Pose graph (`geometry/drift.py`): ARKit odometry edges between consecutive keyframes, plus loop edges at revisits (≥ 100 keyframes apart, < 0.6 m, < 25 deg viewing difference), each verified by coarse-to-fine point-to-plane ICP of local submaps and accepted only if fitness > 0.5 , RMSE < 1.5 cm, correction < 0.3 m and < 3 deg; Open3D global optimisation with line-process pruning. ON by default; `--no-drift-correction` is the ablation switch. A first version (start-end loop only) was rejected by its own checks on every sample (start and end views do not overlap).

Setting (sample with ceiling)	loops used	max pose shift	floor spread	footprint diff vs repeat capture
OFF (ARKit poses)	-	-	1.40 cm	6.4%
ON, strict (shipped)	1/5	16.6 cm	1.49 cm	12.0%
ON, loose (fitness > 0.35)	5/5	19.4 cm	1.80 cm	16.2%

On ARKitScenes capture a it accepted 2/3 loops (max shift 9.1 cm). **No measured benefit on these captures:** ARKit already relocalises on revisits, and partial-overlap loop edges add more error than they remove. Shipped strict so that genuinely drifting captures are corrected; this row is not a pass.

6. Fix loop (25%)

Declared before any fix code (tag `fixloop-before`): worst measured gate = LiDAR repeatability, 0/22 walls; hypothesis: room segmentation (not sensing, drift or snapping); prediction $\geq 70\%$ after structure-only segmentation. Shipped (tag `fixloop-after`): structure-only segmentation, protocol-covered wall band, wall extension, fine yaw, rectangle-first snapping. Result **2/20 walls; matched-room extents median 50 -> 19 cm; best room 2.3/2.6 cm. Gate not passed; prediction badly wrong.** Root cause partly right: segmentation was the dominant cause, but snapping was wrongly excluded and a 0.75 deg orientation error was missed. The laser ground truth found later confirms the snapping half: wrong-face picks of ~ 20 cm. Full post-mortem: `fixloop/POSTMORTEM.md`.

7. Error budget and calibration

Source (LiDAR)	Size	Evidence
Sensor + poses (cloud vs laser)	0.8% scale, 1.9 cm RMSE	<code>scale_check.py</code>
Raw surface repeatability	0.69-0.95 cm	repeatability registrations
Orientation between captures	0.22 deg after refinement	registration residual
Room-side face selection, physical walls	~ 5 cm median (clean-GT walls)	<code>laser_gt.py</code> , clean subset
Virtual walls (doorway closure / extension)	~ 20 cm where they occur	<code>laser_gt.py --peaks</code>
Partition differences between captures	15-140 cm when they occur	<code>fixloop/after/overlay_A_blue_B_red.png</code>

Intervals are fitted from measured errors (`scripts/fit_calibration.py`): LiDAR half-width = $\max(4.3\% \times \text{length}, 46 \text{ cm})$ from 22 laser residuals (errors are roughly constant in cm, so a relative-only interval was 45% covered); video 97% and photo 39% from errors vs LiDAR. Coverage after fitting: LiDAR 91% vs laser, video/photo 100% vs LiDAR. **These coverages are in-sample (fitted and evaluated on the same captures) and therefore optimistic.**

8. Known failure modes and what is not done

- Virtual walls (doorway closure / wall extension) and capture-dependent partitions (wardrobes/shelves in the wall band, sliver rooms) dominate the LiDAR error (~ 20 cm where they occur); on physical walls the error is ~ 5 cm. Free-space face selection was tried and regressed. Next steps: never snap a virtual side to a stray surface (keep it at the doorway-closure line and widen its interval), classify wall vs tall furniture by vertical extent, merge slivers.
- Mirrors and glass: LiDAR sees through or reflects, creating phantom space; not handled.
- Low texture / long walks: video chunk chaining drifts; needs a global pose graph over chunk overlaps.
- Not implemented: windows, damage regions, concealed-damage rules, scope line items (schema fields are emitted empty), consumer-app head-to-head (no device), staged-damage benchmark room.